

Improving the Bumper-Beam Crash Surrogate with PINNeAPLe: Physics-Attention Transolver, Node-wise Scaling and Classical Baselines

From a weekend neural operator to a validated, uncertainty-aware surrogate

PINNeAPLe Labs

PINNeAPLe Labs — Physics AI research notes

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Abstract

Starting from our reproduction of a weekend Transolver for OpenRadioss bumper-beam data (node-1806 RMSE 0.968 mm on the ten held-out runs), we improve the surrogate with PINNeAPLe: (i) the full Physics-Attention Transolver (per-head slice weights, normalised slice tokens, learnable temperature), added natively to the library; (ii) node-wise output scaling, so that small-amplitude structural nodes are not drowned by large rigid motions; (iii) per-node physical features (own gauge and material) instead of broadcast gauges; (iv) the exact initial condition $u(t = 0) = 0$ imposed by construction; (v) a three-member deep ensemble for uncertainty. The improved model reaches a node-1806 RMSE of 0.463 mm ($R^2 = 0.989$) and a field relative L2 of 0.018. We also report what the write-up did not: a classical POD + Gaussian-process surrogate on the two design variables reaches 0.623 mm, and a leave-one-out study over all 100 runs gives its error distribution. The best model on this held-out set is the PINNeAPLe Transolver ensemble.

Keywords: OpenRadioss; crash surrogate; Transolver; physics attention; Gaussian process; POD; uncertainty; PINNeAPLe; VeriPhysics

Main findings

- Node-wise output scaling removes the scale mismatch identified in the reproduction; its effect was not ablated separately from the architecture change.
- With two design variables and 100 runs, a classical POD + GP surrogate is a baseline any neural operator must be compared with.
- Mesh-native neural operators earn their cost when geometry or topology varies — not in this two-gauge study, where only thicknesses change.

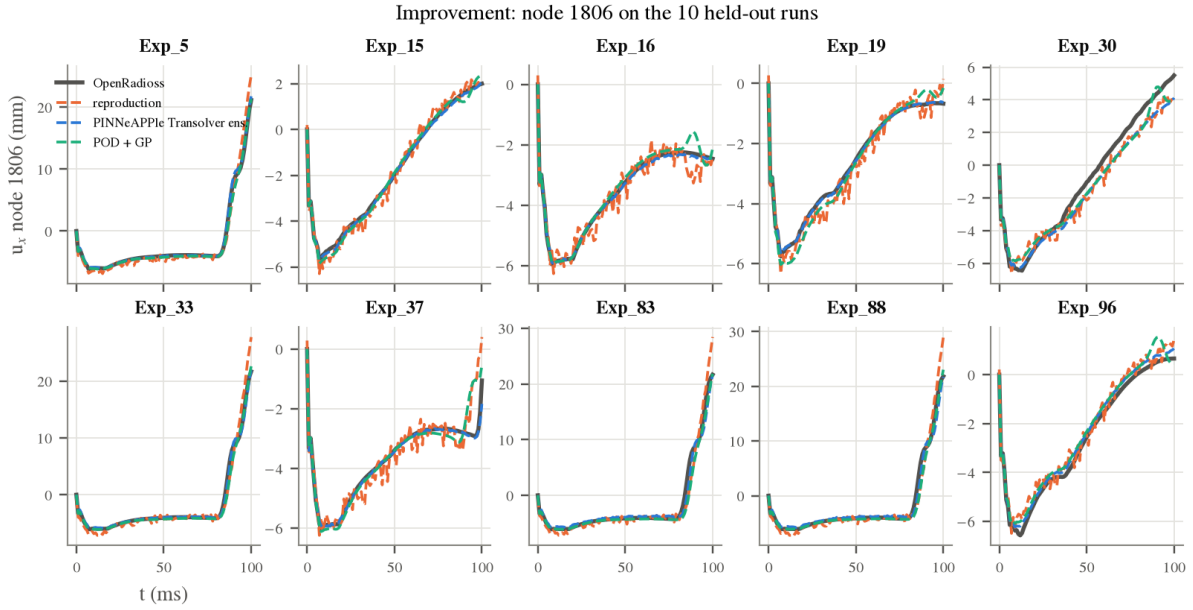
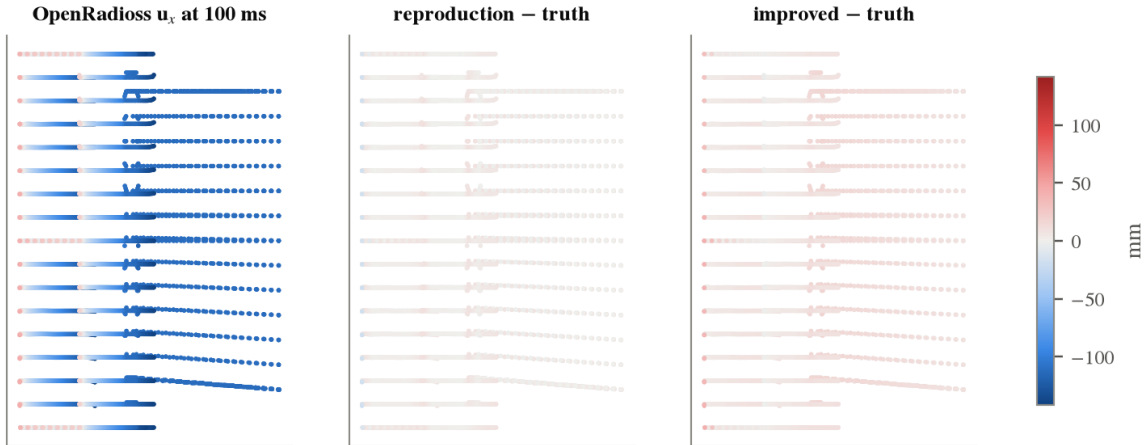
1 Improvements

- **Physics-Attention Transolver** (`pinneapple_neural...transolver.Transolver`, registered as `transolver_native`): 4 blocks, width 128, 8 heads $\times$ 32 slices, MLP ratio 2; pure PyTorch, no research-only dependency (the existing Noether bridge is licence-restricted).
- **Node-wise normalisation** of u_x with training-set mean and standard deviation per node.
- **Features:** normalised coordinates, the node's own shell gauge (0 for non-shell/rigid nodes), material one-hot (DP600 / DP1000 / other), both design variables.
- **Hard initial condition:** the first frame is fixed to zero displacement.
- **Deep ensemble** (3 seeds) for predictive uncertainty; AdamW, cosine schedule, early stopping on the same validation runs.
- **Classical baselines:** POD of the training fields (99.99% energy) with a Matérn-ARD Gaussian process per coefficient; nearest design point.

2 Results

Table 1. Held-out test runs (write-up IDs), u_x .

Model	node RMSE (mm)	node max lerrl (mm)	node R^2	peak err (mm)	final err (mm)	field rel. L2	shell rel. L2
Reproduction (TransolverLite)	0.968	7.163	0.952	0.593	2.844	0.033	0.028
PINNeAPPLE Transolver ensemble	0.463	3.179	0.989	0.162	0.381	0.018	0.017
POD + Gaussian process	0.623	5.188	0.980	0.166	0.592	0.005	0.006
Nearest design point	0.445	2.522	0.990	0.097	0.605	0.022	0.022

**Figure 1.** Node-1806 history on the ten held-out runs: OpenRadioss vs reproduction, improved ensemble and POD + GP.**Figure 2.** u_x at 100 ms on the shell structure (x-z view) for one test run, and the errors of the two neural models.

Leave-one-out over all 100 runs (POD + GP, node 1806): median RMSE 0.325 mm, 95th percentile 0.772 mm, maximum 0.847 mm — a distribution, not one split.

2.1 Uncertainty and VeriPhysics

Table 2. PINNeAPPLE `compute_physics_confidence` against the held-out node-1806 histories (registered benchmark). `uq_calibration` uses the ensemble spread (Transolver) or the GP posterior (POD + GP). Not run: `physics_guardrail` (no compiled PDE ProblemSpec exists for explicit shell-FE crash dynamics); `numerical_convergence` (single mesh supplied by the example; no refinement study was run); `geometry_ood` (geometry is identical across the DoE (only gauges vary)).

Model	Trust score	Coverage	Components
Reproduction: TransolverLite (write-up)	0.806	0.2	benchmark_agreement 0.81
Improved: PINNeAPPLE Transolver ensemble	0.907	0.4	uq_calibration 0.91; benchmark_agreement 0.91
POD + Gaussian process (DV1, DV2)	0.891	0.4	uq_calibration 0.91; benchmark_agreement 0.88
Nearest design point	0.911	0.2	benchmark_agreement 0.91

3 Discussion

The design space here is two-dimensional and the mesh never changes, which is the regime where interpolation in design space (POD + GP) is hard to beat. A mesh-native operator is justified when the geometry varies between designs or when fields must be predicted for unseen meshes; that is the natural next experiment, together with more runs near the design-space edges where all models degrade.

References

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